Xinyuan Song

Ph.D. Student in Computer Science · Emory University · Atlanta, Georgia

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Google Scholar: 2,507 citations · h-index 24 · i10-index 51 (September 2026).

Education

Emory University 2024 – Present
Ph.D. in Computer Science; focus on Machine Learning. Advisor: Liang Zhao.

University of Texas at Arlington 2023 – 2024
Ph.D. studies in Computer Science. Advisor: Yingying Zhu.

National University of Singapore 2019 – 2023
M.S. in Statistics. Advisor: Cheng Li.

Peking University 2014 – 2019
B.S. in Physics; B.S. in Mathematics.

Industry Experience

Cielara AI Summer 2026
Research on memory for language-model agents. MEMONDEMAND develops query-adaptive hierarchical memory for large enterprise collections; MegaMem provides source-resolved retrieval and attribution over ultra-large context windows.

Gradient Network Summer 2025
Research on multi-agent systems, including GradientSys: A Multi-Agent LLM Scheduler with ReAct Orchestration.

Alibaba Group 2020 – 2021
Full-time Machine Learning Engineer.

Selected Publications

1. CPAL · 2026 ORAL
   MASTERPIECE **Matrix Sensing with Kernel Optimal Loss: Robustness and Optimization Landscape**
   Xinyuan Song and Ziye Ma.
   PDF GITHUB BIBTEX
2. NeurIPS · 2025
   The Curse of Depth in Large Language Models
   Wenfang Sun*, **Xinyuan Song***, Pengxiang Li, Lu Yin, Yefeng Zheng, and Shiwei Liu.
   PDF GITHUB BIBTEX
3. NeurIPS · 2025
   TRiCo: Triadic Game-Theoretic Co-Training for Robust Semi-Supervised Learning
   Hongyang He*, **Xinyuan Song***, Yangfan He, Zeyu Zhang, Yanshu Li*, Haochen You, Lifan Sun, and Wenqiao Zhang.
   PDF BIBTEX
4. ICML · 2026
   Newton-Coupled Dual-Teacher Semi-Supervised Learning Framework
   Hongyang He*, **Xinyuan Song***, Yan Zhong, Daizong Liu, Xuanyu Liu, and Victor Sanchez.
   PDF BIBTEX
5. ICML · 2026
   When Does Sparsity Mitigate the Curse of Depth in LLMs
   Dilxat Muhtar*, **Xinyuan Song***, Sebastian Pokutta, Max Zimmer, Nico Pelleriti, Thomas Hofmann, and Shiwei Liu.
   PDF GITHUB BIBTEX
6. ICASSP · 2026
   Demystifying the Roles of LLM Layers in Retrieval, Knowledge, and Reasoning
   Xinyuan Song, Keyu Wang, Pengxiang Li, Lu Yin, and Shiwei Liu.
   PDF GITHUB BIBTEX

7. CPAL · 2026
Efficient Temporal Consistency in Diffusion-Based Video Editing with Adaptor Modules: A Theoretical Framework
Xinyuan Song, Yangfan He, Sida Li, Jianhui Wang, Hongyang He, Xinhang Yuan, Ruoyu Wang, Jiaqi Chen, Keqin Li, Kuan Lu, Menghao Huo, Binxu Li, and Pei Liu.
PDF BIBTEX
8. ACL Findings · 2026
P-QuASAR: A Unified Probabilistic Framework for Holistic Patent Quality Assessment and Refinement
Xinyuan Song, Ziyi Ni, Fred Yang, Bo Zhang, Yiming Yan, Yijin Wang, and Jane Zhang.
PDF BIBTEX
9. CPAL · 2026
Token-Aware Representation Augmentation for Fine-Grained Semi-Supervised Learning
Hongyang He*, Yan Zhong*, **Xinyuan Song***, Daizong Liu, and Victor Sanchez.
PDF BIBTEX
10. ECCV · 2026
Geometric Regularization for Long-Tailed Semi-Supervised Learning via Gaussian Feature Bridges
Hongyang He*, **Xinyuan Song***, Yan Zhong, Daizong Liu, Yanbin Li, Yangfan He, and Wenqiao Zhang.
PDF BIBTEX
11. ACM Multimedia · 2025
Free-Mask: A Novel Paradigm of Integration Between the Segmentation Diffusion Model and Image Editing to Improve Segmentation Ability
Bo Gao*, Jianhui Wang*, **Xinyuan Song***, Yangfan He, Fangxu Xing, and Tianyu Shi.
PDF BIBTEX
12. Heliyon · 2023
Predicting Stock Price of Construction Companies Using Weighted Ensemble Learning
Xinyuan Song.
PDF BIBTEX

Recent Study

1. arXiv 2609.17419 · 2026
World Model Science: Self-Organized Criticality, Weak Chaos, and Metastable Belief Dynamics in Long-Horizon LLM Agents
Xinyuan Song and Zekun Cai.
PDF GITHUB BIBTEX
2. arXiv 2608.22141 · 2026
MEMONDEMAND: A Memory Management System for Large-Scale Enterprise Data
Xinyuan Song, Bowen Zhu, Hasibul Haque, and Liang Zhao.
PDF GITHUB BIBTEX
3. arXiv 2608.22137 · 2026
MegaMem: A Retrieval Solution for Ultra-Large Context Windows
Xinyuan Song, Bowen Zhu, Hasibul Haque, and Liang Zhao.
PDF GITHUB BIBTEX
4. arXiv 2607.19331 · 2026
ISO: An RLVR-Native Optimization Stack
Hanqing Zhu, Wenyan Cong, Zhizhou Sha, Sagnik Mukherjee, **Xinyuan Song**, David González-Martínez, Xiaoxia Wu, Yuandong Tian, Shiwei Liu, David Z. Pan, and Zhangyang “Atlas” Wang.
PDF GITHUB BIBTEX
5. arXiv · 2026
FedWorld: Scope-Aware Federation of Agent World Models
Yuchao Hou, Guangyuan Dong, **Xinyuan Song**, Hangtao Zhang, Gaoyuan Du, Jiachen Luo, and Zhengshan Wang.
PDF GITHUB BIBTEX
6. arXiv · 2026
When Do Safe Agents Misgeneralize Across Surfaces? Invariant Safety State Abstraction
Yuchao Hou, Guangyuan Dong, Hangtao Zhang, **Xinyuan Song**, Jiachen Luo, Gaoyuan Du, and Xianlong Wang.
PDF GITHUB BIBTEX
7. arXiv 2607.05458 · 2026
Learning to Control LLM Agent Harnesses with Offline Reinforcement Learning
Haiwen Yi and **Xinyuan Song**.
PDF GITHUB BIBTEX
8. arXiv 2607.04535 · 2026
ManifoldFlow: SPD-Relaxed Stiefel Layers with Learnable Singular Spectrum
Haiwen Yi and **Xinyuan Song**.
PDF GITHUB BIBTEX

9. **arXiv 2607.04528 · 2026**
   Measuring Harness-Induced Belief Divergence in Multi-Step LLM Agents
   Haiwen Yi and **Xinyuan Song**.
   PDF GITHUB BIBTEX
10. **arXiv 2607.01767 · 2026**
    Repair the Amplifier, Not the Symptom: Stable World-Model Correction for Agent Rollouts
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
11. **arXiv 2607.00062 · 2026**
    AlgoBench: Benchmarking Algorithmic Adaptation in Code Generation
    Xinyuan Song, Zekun Cai, and Liang Zhao.
    PDF GITHUB BIBTEX
12. **arXiv 2606.31422 · 2026**
    Ask the World Before Acting: Environment Probing for Calibrated Agent World Models
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
13. **arXiv 2606.31399 · 2026**
    World-Model Collapse as a Phase Transition
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
14. **arXiv 2606.29999 · 2026**
    AlgoSkill: Learning to Design Algorithms by Scheduling Human-Like Skills
    Xinyuan Song, Zekun Cai, and Liang Zhao.
    PDF GITHUB BIBTEX
15. **arXiv 2606.28438 · 2026**
    When AI Reviews Its Own Code: Recursive Self-Training Collapse in Code LLMs
    Xinyuan Song, Zekun Cai, and Liang Zhao.
    PDF GITHUB BIBTEX
16. **arXiv 2606.27806 · 2026**
    Agent vs. Parametric World Models: Hybrid Planning for Reliable Language Agents
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
17. **arXiv 2606.27780 · 2026**
    Understanding Rollout Error in Graph World Models
    Xinyuan Song and Zekun Cai.
    PDF GITHUB BIBTEX
18. **arXiv 2605.13716 · 2026**
    SkillOps: Managing LLM Agent Skill Libraries as Self-Maintaining Software Ecosystems
    Xinyuan Song, Hongji Pu, and Liang Zhao.
    PDF GITHUB BIBTEX

Collaborative Publications

1. **IEEE TMI · 2026**
   Med-MARS: Modifier-Aware Referring Segmentation for Medical Images
   Yajun Mu, Huahui Yi, Jun Gao, Ziyuan Qin, **Xinyuan Song**, Qicheng Lao, Qingbo Kang, and Kang Li.
   PDF BIBTEX
2. **NeurIPS · 2026**
   Harder the Task, Sparser the Representation: Sparsity as a Learning Signature of Capability in LLMs
   Mingyu Jin, Yutong Yin, Jingcheng Niu, Qingcheng Zeng, Wujiang Xu, **Xinyuan Song**, Wei Cheng, Mengnan Du, Zhaoran Wang, Tianlong Chen, and Dimitris N. Metaxas.
   PDF GITHUB BIBTEX
3. **NeurIPS · 2026**
   CONSTRAINER: Promptable Graph-Structured Optimization via Constraint Conditioning
   Zeeshan Memon, Mingke Tian, **Xinyuan Song**, Hongwei Jin, Kibaek Kim, and Liang Zhao.
4. **ACL Findings · 2026**
   EffectCodec: Emotion-Preserving Neural Speech Codec for Expressive Speech Modeling
   Jiacheng Shi, Hongfei Du, **Xinyuan Song**, Y. Alicia Hong, Yanfu Zhang, and Ashley Gao.
   PDF BIBTEX
5. **CVPR · 2026**
   DICArt: Advancing Category-level Articulated Object Pose Estimation in Discrete State-Spaces
   Li Zhang, Mingyu Mei, Ailing Wang, Xianhui Meng, Yan Zhong, **Xinyuan Song**, Liu Liu, Rujing Wang, Zaixing He, and Cewu Lu.
   PDF BIBTEX

6. ACM Multimedia · 2026
Generalizable Blind Image Quality Assessment with Retrieval Augmentation and Contrastive Vision-Text Local Meta-Training
Yan Zhong, Zhiyuan You, Xinping Zhao, Li Zhang, **Xinyuan Song**, Ruoyu Zhao, Lei Shi, and Tingting Jiang.
PDF BIBTEX

7. CPAL · 2026
ShapLoRA: Allocation of Low-rank Adaptation on Large Language Models via Shapley Value Inspired Importance Estimation
Colin Zhao, Qinghua Yao, **Xinyuan Song**, and Wei Zhu.
PDF BIBTEX

8. CPAL · 2026
Enhancing Low-Cost Video Editing with Lightweight Adaptors and Temporal-Aware Inversion
Yangfan He, Sida Li, Jianhui Wang, Kun Li, **Xinyuan Song**, Xinhang Yuan, Keqin Li, Kuan Lu, Menghao Huo, Jingqun Tang, Yi Xin, Jiaqi Chen, Miao Zhang, and Xueqian Wang.
PDF BIBTEX

9. CPAL · 2026
Improving Medical Visual Reinforcement Fine-Tuning via Perception and Reasoning Augmentation
Guangjing Yang, ZhangYuan Yu, Ziyuan Qin, **Xinyuan Song**, Huahui Yi, Qingbo Kang, Jun Gao, Yiyue Li, Chenlin Du, and Qicheng Lao.
PDF BIBTEX

10. ICASSP · 2026
Optimizing Multi-Round Enhanced Training in Diffusion Models for Improved Preference Understanding
Kun Li, Jianhui Wang, Yangfan He, **Xinyuan Song**, Ruoyu Wang, Hongyang He, Wenxin Zhang, Jiaqi Chen, Keqin Li, Sida Li, Miao Zhang, Tianyu Shi, and Xueqian Wang.
PDF BIBTEX

11. ICASSP · 2026
Self-evolving Agents with Reflective and Memory-Augmented Abilities
Xuechen Liang, Yangfan He, Yinghui Xia, **Xinyuan Song**, Jianhui Wang, Meiling Tao, Li Sun, Xinhang Yuan, Jiayi Su, Keqin Li, Jiaqi Chen, Jinsong Yang, Siyuan Chen, and Tianyu Shi.
PDF GITHUB BIBTEX

12. ICLR CAO Workshop · 2026
Manifold-Aware Temporal Domain Generalization for Large Language Models
Yiheng Yao, Zekun Cai, **Xinyuan Song**, Hiroki Hill Kobayashi, Xuan Song, Ryosuke Shibasaki, and Liang Zhao.
PDF BIBTEX

13. ICLR LLM Reasoning Workshop · 2026
Enhancing Code LLMs with Reinforcement Learning in Code Generation: A Survey
Junqiao Wang, Zeng Zhang, Yangfan He, Zihao Zhang, **Xinyuan Song**, Yuyang Song, Tianyu Shi, Yuchen Li, Hengyuan Xu, Kunyu Wu, Xin Yi, Zhongwei Wan, Xinhang Yuan, Zijun Wang, Kuan Lu, Menghao Huo, Tang Jingqun, Guangwu Qian, Keqin Li, Qiuwu Chen, and Lewei He.
PDF BIBTEX

14. WWW LARS Workshop · 2026
SCORE: Story Coherence and Retrieval Enhancement for AI Narratives
Qiang Yi, Yangfan He, Jianhui Wang, **Xinyuan Song**, ShiYao Qian, Xinhang Yuan, Yi Xin, Yijin Wang, Jingqun Tang, Yuchen Li, Junjiang Lin, Hongyang He, Zhen Tian, Tianxiang Xu, Keqin Li, Kuan Lu, Menghao Huo, Jiaqi Chen, Miao Zhang, Tianyu Shi, and Jianyuan Ni.
PDF BIBTEX

15. CVPR DataCV Workshop · 2026
Beyond Data Augmentation: Energy-Based Kuramoto Neurons for Semi-Supervised Learning
Yundi Hong, Ao Li, **Xinyuan Song**, Yan Zhong, Hongyang He, and Victor Sanchez.
PDF BIBTEX

16. BIO Integration · 2026
Large Language Models and Cognitive Science: A Comprehensive Review of Similarities, Differences, and Challenges
Qian Niu, Junyu Liu, Ziqian Bi, Pohsun Feng, Benji Peng, Keyu Chen, Ming Li, Lawrence KQ Yan, Yichao Zhang, Caitlyn Heqi Yin, Cheng Fei, Tianyang Wang, Yunze Wang, Silin Chen, Ming Liu, Ziyuan Qin, Riyang Bao, **Xinyuan Song**, and Zekun Jiang.
PDF BIBTEX

17. NeurIPS · 2025
GPAS: Accelerating Convergence of LLM Pretraining via Gradient-Preserving Activation Scaling
Tianhao Chen, Xin Xu, Zijing Liu, Pengxiang Li, **Xinyuan Song**, Ajay Kumar Jaiswal, Fan Zhang, Jishan Hu, Yang Wang, Hao Chen, Shizhe Diao, Shiwei Liu, Yu Li, Lu Yin, and Can Yang.
PDF GITHUB BIBTEX

18. TKDD · 2025
Bridging NIP and MLM: A Unified Meta-Learning Framework for Personalized Sequential Recommendation
Youhua Li, Ersheng Ni, Yihao Liu, Sibo Xu, Tianyi Xu, Mingxuan Wu, Junchen Fu, Yucheng Zhang, Yuanqi He, **Xinyuan Song**, and Yongxin Ni.
PDF BIBTEX

19. EMNLP · 2025
SOLAR: Serendipity Optimized Language Model Aligned for Recommendation
Zichen Yuan, Lifan Sun, Yucen Zhuang, Yue Wang, **Xinyuan Song**, Tianqi Xu, Siyuan Li, Junchen Fu, Youhua Li, Sirui Hong, Jiaqi Chen, Joemon M. Jose, and Yongxin Ni.
PDF GITHUB BIBTEX

20. CIKM · 2025
AMBER: Adaptive Meta Balanced Paradigm for Heterogeneous Graph-Based Knowledge Tracing
Lifan Sun, Zichen Yuan, Ersheng Ni, Weihua Cheng, **Xinyuan Song**, Linkun Dai, Hongwei Jiang, Sibo Xu, Mengmeng Chen, Yucen Zhuang, Yongxin Ni, and Youhua Li.
PDF BIBTEX

21. ACM Multimedia · 2025
Adaptive Prompt Learning for Blind Image Quality Assessment with Multi-modal Mixed-datasets Training
Yan Zhong, Xinping Zhao, Li Zhang, **Xinyuan Song**, and Tingting Jiang.
PDF BIBTEX

22. ICME · 2025
ContrastiveGaussian: High-Fidelity 3D Generation with Contrastive Learning and Gaussian Splatting
Junbang Liu, Enpei Huang, Dongxing Mao, Hui Zhang, **Xinyuan Song**, and Yongxin Ni.
PDF GITHUB BIBTEX

23. IEEE RA-L · 2025
Towards the Integrated Category-level Articulation Perception and Manipulation Framework for Embodied Intelligence
Li Zhang, Wenbo Xu, Jiankun Wang, **Xinyuan Song**, Qi Wu, Rujing Wang, Xue Wang, and Liu Liu.

24. Neurocomputing · 2025
Enhancing Intent Understanding for Ambiguous Prompt: A Human-Machine Co-adaption Strategy
Yangfan He, Jianhui Wang, Yijin Wang, Yan Zhong, **Xinyuan Song**, Junjiang Lin, Xinhang Yuan, Jingqun Tang, Yi Xin, Hao Zhang, Yuchen Li, Zijian Zhang, Hongyang He, Tianxiang Xu, Miao Zhang, Kuan Lu, Menghao Huo, Keqin Li, Jiaqi Chen, Tianyu Shi, and Jianyuan Ni.
PDF BIBTEX

25. Transportation Research Record · 2025
Who Can You Trust on the Road? Adversarial Intentions in Cooperative MARL
Siyang Zhang, Yecheng Lyu, Luyang Gong, **Xinyuan Song**, and Tianyu Shi.
PDF BIBTEX

26. BIBM · 2025
Multi-Scale Volumetric Transformers with Adaptive Uncertainty Modeling for Robust Bacterial Flagellar Motor Localization in Cryo-Electron Tomography
Tianxiang Xu, Junqiao Wang, Yangfan He, **Xinyuan Song**, Peilai Yu, Xunfei Zhu, and Chang Liu.
PDF BIBTEX

27. BIO Integration · 2025
From Bench to Bedside: A Review of Clinical Trials in Drug Discovery and Development
Tianyang Wang, Ming Liu, Benji Peng, **Xinyuan Song**, Charles Zhang, Xintian Sun, Qian Niu, Junyu Liu, Silin Chen, Keyu Chen, Ming Li, Pohsun Feng, Ziqian Bi, Yunze Wang, Yichao Zhang, Cheng Fei, Lawrence KQ Yan, Ziyuan Qin, Riyang Bao, and Zekun Jiang.
PDF BIBTEX

28. Neural Networks · 2024
Semi-Supervised Multi-Label Feature Selection with Consistent Sparse Graph Learning
Yan Zhong, Xingyu Wu, Xinping Zhao, Li Zhang, **Xinyuan Song**, Lei Shi, and Bingbing Jiang.
PDF BIBTEX

* Equal contribution.

Honors and Awards

- Silver Medal, Chinese Physics Olympiad (national level)
- Silver Medal, Chinese Mathematics Olympiad (national level)